

Article

A Comparative Study between Paper and Paperless Aircraft Maintenance: A Case Study

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Abstract: The rapid growth of airlines over the last period has led to the development of the aviation industry, resulting in increased requirements for ancillary services. Nowadays, the demand for the use of paper-based documents is increasing day by day as airlines encourage passengers to use more ancillary products. One of the most important issues in the aviation industry is flight safety. The procedures and instructions required for all aviation operations are organized in accordance with the regulations and printed on paper in accordance with the demands. For this reason, since it is mandatory to keep all aircraft maintenance, repair, and operation records on printed paper and to keep them for a certain period of time, aircraft fly with almost their entire weight on paper. Considering the global hazards in the world, although the aviation sector is the least environmentally damaging of all transportation sectors, new solutions have been sought to make aviation operations less harmful to the environment, minimize errors and risks, allow for faster communication, and be safer and more sustainable. In order to meet the demands, studies on reducing the use of paper have been brought to the agenda. In this study, after a literature review, real data from a maintenance, repair, and overhaul organization are used to suggest digital conveniences that can save costs, increase productivity, save space, facilitate documentation and information sharing, keep personal information more secure, and help the environment through paperless living.

Keywords: aviation operations; digital transformation in aviation; paperless aviation; paperless aircraft maintenance; sustainability in aircraft maintenance



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1. Introduction

It is thought that paper was first produced around 105 BC by Cai Lun by crushing softened tree bark, hemp waste, old rags, and fishing nets until they became soft and pulpy, and then mixing the resulting pulp with water to make wood pulp. The paper has been used in many fields from the past to the present, ranging from industry to the aviation sector [1–3]. Aircraft operations and maintenance are heavily regulated and meticulously planned to ensure safety [4]. Regulations require a large amount of paperwork as they are followed by legal processes. Paperwork is handled by multiple stakeholders, including Original Equipment Manufacturers (OEMs) [5], leaseholders [6], Maintenance Repair and Overhaul Organizations (MROs) [7], suppliers, and airlines [8]. It is imperative for each party involved to efficiently handle their transactions and data in accordance with the regulations set forth by the registration authority. The existing process for aircraft operations

and maintenance is currently undergoing a gradual transformation through the implementation of digitalization [9]. As stakeholders embrace novel technological solutions, it is anticipated that the quantity of information produced will escalate, consequently augmenting its inherent value. The implementation of key enabling solutions and initiatives, such as digital aircraft (commonly known as paperless aircraft), operations and maintenance, e-signature, and Radio Frequency Identification (RFID), gives rise to further challenges [10]. In addition, it is imperative to establish consistency among technological solutions through the adoption of universal requirements and unified information standards. The International Air Transport Association (IATA) has developed an implementation guide for paperless aircraft operations with the aim of providing support to both stakeholders and scholars [11,12]. Throughout the life cycle of a commercial aircraft, some activities, such as repairs, replacement of components, and maintenance tasks, are defined on a day to day basis, creating records that need to be reconciled. Leaseholders have strict contractual terms requiring operators to retain and make all necessary maintenance documentation available to preserve asset value. Since all contractual transactions are nowadays mostly carried out on paper, the transaction cost is quite high [13–16].

Paper production generates greenhouse gas emissions, which lead to deforestation and anthropogenic climate change [17]. According to the American Forest and Paper Association, paper production is the third largest fossil fuel-consuming process worldwide [18]. According to a study from 2001, only 50 companies account for 43% of the world's annual industrial wood harvesting volume [19]. On the other hand, the lack of discussions on forest management and its relationship to land use, resource management, and biodiversity was identified as a particular weakness in the report prepared as part of the study on the Canadian Forest and Paper Industry [20]. The benefits of a paperless environment, also known as digitalization, can be briefly summarized as reducing the carbon footprint, benefiting wildlife, providing accessibility to documents from anywhere, and easing storage [18,21].

Within the scope of this study, the impact of implementing digital maintenance records to replace paper documents in commercial aviation is explored by examining the benefits of increased flight efficiency and cost reduction.

The research makes several significant contributions to the understanding of paperless practices in the aviation industry, particularly in the context of aircraft maintenance, repair, and overhaul. These contributions can be summarized as follows: (i) Environmental awareness: The paper highlights the environmental implications of the aviation industry's heavy reliance on paper-based documentation. It underlines the industry's role in seeking sustainable solutions to reduce resource consumption, thus addressing environmental concerns such as deforestation and greenhouse gas emissions. (ii) Operational efficiency: It emphasizes the potential for increased operational efficiency resulting from the adoption of paperless practices. By reducing their reliance on printed materials and transitioning to digital tools like tablets, organizations can streamline processes, reduce costs, and improve resource utilization. (iii) Technological integration: By incorporating digitalization into their operations, airline companies could aim to optimize the utilization of resources and enhance overall operational precision. (iv) Cost and resource savings: The research provides concrete data on the cost savings associated with transitioning to paperless maintenance, including reductions in paper usage, storage space, and energy consumption. This information quantifies the economic benefits of paperless practices. (v) Real-world case study: It presents a real-world case study of a private aircraft maintenance organization in Turkey that has successfully implemented paperless practices. This case study serves as an illustrative example of the tangible benefits and savings achievable through digitalization.

2. Industry 3.0 and 4.0—Paper

Nowadays, in companies operating in the economic and industrial sectors, specializing in the development and production of different items, and using Industry 3.0, there are some specific paper and electronic document technologies. These technologies are

built upon the development of Industry 3.0 companies, the components of the six stages of modernization, and the application of these technologies in project and production company environments [22,23]. Modernization in Industry 3.0 companies can be briefly summarized in six stages [24]:

1. Providing the company with digital projection and production tools (company computerization).
2. A digital projection of company modernization and implementation of new technologies that combine the means of production with the digital means of operations (paperlessness).
3. Implementation of hybrid technological lines in smart factories with unmanned and paperless production.
4. Utilizing augmented reality technology (Big Data), the implementation of data collection and processing aids company personnel in effectively managing and monitoring the progress of projects and production operations.
5. Application of some components and technologies to predict activity.
6. Self-regulation of technological equipment (company adaptability) through artificial intelligence technology.

Within the context of smart factories, there are specific algorithms designed to calculate the mean time between failures (MTBF) for production infrastructure. These phenomena are derived from the self-organization of both cybernetic and physical systems within the context of technological frameworks [25].

In recent years, there has been a significant transformation in maintenance functions due to the rapid advancement of technology. In accordance with the study conducted by Ahmad and Kamaruddin [26], maintenance is characterized as a collection of activities or tasks employed to reinstate an object to a condition wherein it is capable of executing its intended functions. The notion of preventive maintenance entails the execution of maintenance tasks prior to the occurrence of equipment failure. One of the primary objectives of preventive maintenance is to mitigate the rate or frequency of equipment failure. The implementation of a preventive maintenance strategy plays a significant role in reducing costs associated with equipment breakdowns and minimizing machine downtime, thereby enhancing overall product quality [27]. The strategy is also becoming particularly evident at the Manufacturing Operations Management (MOM) level [28], with preventive maintenance playing a role across all levels of manufacturing operations. According to the findings of Savage [29], Manufacturing Execution Systems (MES) encompass a collection of technologies that are implemented at the Manufacturing Operations Management (MOM) level. The successful implementation of MES in the pharmaceutical industry has been observed following the issuance of the final 21 Part 11 regulations by the Food and Drug Administration (FDA) on 21 March 1997 [30]. The regulations provided various criteria for acceptance of electronic records by the FDA under certain conditions (such as electronic signatures, handwritten signatures executed on paper, and handwritten signatures executed on electronic records as equivalent to paper records) [31]. For over 16 years, the technology associated with MES has evolved over time, rapidly becoming one of the best practices recognized in all pharmaceutical and life sciences regulated industries [32]. This is validated by the fact that most greenfield production facilities start with an MES in place, enabling essentially paperless production from day one [33]. The establishment of future-oriented enterprises that rely on paperless and unmanned technologies is emerging as a critical challenge in the industry, necessitating a global-scale resolution. In the present era, it is not feasible for any nation to autonomously develop the complete life cycle of a product, exclusively relying on its domestic technologies and components. The phenomenon of internationalization of production has given rise to the formation of transnational corporations, characterized by a division of tasks across multiple countries. In this arrangement, one country serves as the base for marketing and sales activities, another country is responsible for the actual production of the product, and the consumers of the product can be situated

in a third country [34]. At the time of the fourth industrial revolution, the virtual factories of Logistics Industry 4.0 were being built [25,35].

At the Hannover Expo in 2011, the German government introduced the term “Industry 4.0” to describe a strategic approach to manufacturing based on the computerization of production, as envisioned by politics [36]. In the context of the data-driven and intelligence-focused advancement of Industry 4.0, it is anticipated that artificial intelligence techniques, such as machine learning and deep learning, will assume considerable significance in the coming years for the purpose of processing and leveraging the data generated within the production setting [37]. Furthermore, alongside the utilization of artificial intelligence, there is presently an increased discourse and investigation surrounding Industry 4.0, encompassing both scholarly inquiry and managerial literature. The correlation between Industry 4.0 and sustainability, as well as the potential impact of Industry 4.0 on enhancing sustainability, has garnered significant attention in recent academic research [38]. The topic of sustainability is frequently discussed by researchers, who commonly make reference to the widely recognized dimensions of sustainability, namely, economic, environmental, and social sustainability [39]. Numerous scholars assert that Industry 4.0 possesses substantial potential for enhancing productivity, thereby fostering long-term economic sustainability and the capacity to endure. According to Jeske et al. [40], it has been asserted that there should be a projected increase in productivity of 32% by the year 2025. Moreover, additional scholars are currently examining the implications of Industry 4.0 on the aspect of environmental sustainability [41]. Jabbour et al. [42] assert that the implementation of Industry 4.0 technologies has the potential to facilitate green manufacturing, thereby serving as a catalyst for promoting environmental sustainability. Numerous scholars posit that the advent of Industry 4.0 will augment the significance of human involvement in the realm of production, consequently bolstering the aspect of social sustainability [43].

In recent years, the term Industry 4.0 has broadened to include new trends suggesting a revolution built on tangible interaction between production robots, humans, and machines themselves. As the literature suggests [44], the fourth industrial revolution is characterized by four fundamental pillars: interoperability, information transparency, technical assistance, and decentralized decision making.

- Interoperability pertains to the utilization of the Internet of Things (IoT) for the purpose of networking individuals, devices, machines, and robots with the overarching objective of achieving extensive automation in production, commonly known as the “fully automated factory” [45].
- The concept of “information transparency” pertains to the notion of the Digital Twin [46], which involves creating virtual replicas of physical objects and enhancing them with data obtained from actual sensors.
- The introduction of disruptive technologies in Industry 4.0 also encompasses the domain of technical assistance [47]. The term “human-automation collaboration” pertains to the substitution of human labor with cyber-physical machines in order to carry out D-3 operations (dull, dirty, dangerous). These machines offer assistance to operators by providing them with relevant information that can be visualized as required, enabling them to efficiently resolve problems within a limited timeframe.
- Lastly, the concept of decentralized decision making [48] proposes intelligent machines capable of making automated decisions, solving conflicts, and complex planning problems without human intervention, transforming the operator into a supervisor instead of a problem solver.

From a practical standpoint, Industry 4.0 envisions the following [49]:

- The implementation of emerging technologies such as Augmented Reality (AR) and Virtual Reality (VR) within organizations, along with the utilization of Additive Manufacturing (AM), to enhance production efficiency and enable intelligent structures [50];

- By offering software tools that are capable of effectively managing substantial volumes of data, companies can address the challenge commonly referred to as the Big Data problem [51];
- The development of software algorithms that effectively gather data in a clear and user-friendly manner is crucial. This ensures that human auditors are presented with only the most relevant information, preventing information overload caused by excessive data [52].

In light of this, it is believed that the design applications of Industry 4.0 concepts can be adapted from an industrial field to the aviation field by providing maintenance methods, in-flight structural health monitoring, and flight management support. Aviation is a complex and challenging industry in terms of both design and maintenance [53]. For instance, the article by French et al. [54] addresses the challenge of integrating additive manufacturing into aviation, specifically within the context of aviation maintenance. Recent studies in the literature suggest that the concepts of the Industry 4.0 program can be applied to reduce maintenance time and take advantage of the new capabilities provided by technologies such as AM and AR. While numerous studies already exist on these topics, the objective of the innovation presented in this paper is to offer a sustainable progression of maintenance practices by incorporating concepts that are anticipated to be implemented in factories in the coming decade. Figure 1 depicts a chronological representation of the timeline pertaining to the implementation of the technologies elucidated within the document [50].

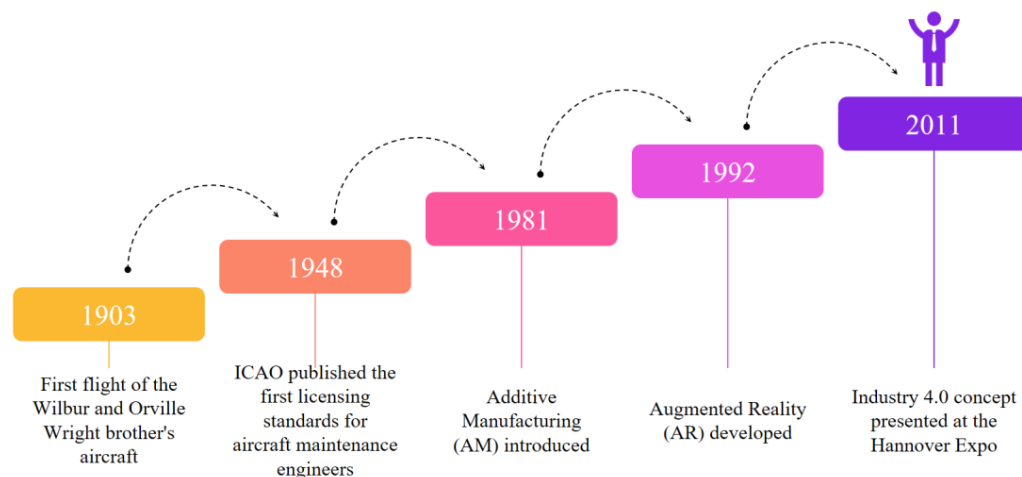


Figure 1. Technology timeline.

3. Aviation Operations—Paper

3.1. Paperlessness in the Green Airports Project

Airports are one of the most important infrastructure elements of civil aviation [55]. Terminal buildings and many supporting equipment in the airspace consume high amounts of energy, and with the significant carbon emissions of aircraft, planning for green airports has become imperative for future airport construction [56]. The first green airport recognized by the Civil Aviation Administration of China (CAAC) is Kunming Changshui International Airport, as evidenced by the case of green airports in China [57]. The CAAC News reported on the 2020 exhibition titled “Four Characteristics of Airport Construction and Development Achievements”, which highlighted six notable projects in Green Airport Construction. The airports in question are Daxing International Airport, Capital Airport, Baiyun Airport, Shangrao Sanqingshan Airport, Changsha Airport, and Chengdu Shuangliu Airport. These airports are identified by their respective IATA Airport Codes, namely, PKX, PEK, CAN, SQD, CSX, and CTU [58]. One of the construction projects at Daxing International Airport includes the implementation of paperless travel initiative and the development of a green airport [16]. The implementation of the paperless travel initiative at Daxing International Airport proposes the feasibility of reducing the quantity of

boarding passes, mitigating paper waste, minimizing CO₂ emissions during the production process, and mitigating environmental pollution. Furthermore, it is worth noting that the aforementioned measures have the potential to enhance the level of security for passengers, expedite the processes of check-in and boarding, optimize the selection of aircraft taxi routes, and effectively manage flight trailers in the airport of Guangzhou, ultimately resulting in improved operational efficiency. Consequently, these advancements enable airports to be categorized as smart airports [59].

3.2. The Place of Paper in Airline Tickets—Electronic Ticketing

Wang and Lv [60] presented statistics demonstrating that revenue from additional services in global airlines reached USD 13.5 billion in 2009, a 43% increase compared to 2008. In 2010, revenue increased by 67% compared to 2009 revenues, which reached USD 22.6 billion. Auxiliary services have generated substantial revenues for airlines and have become a new point of profit growth in the aftermath of the global financial crisis. Compared to pure ticket sales, they noted that ancillary services are rapidly changeable and have a large number of document types. Hence, the consolidation of paper documents into a unified electronic format and the realization of entirely electronic systems in the civil aviation industry have become significant technological challenges [61]. In order to reduce cost ratios, the International Air Transport Association (IATA) has recently started to implement the entire electronic process in the civil aviation industry and to handle all kinds of miscellaneous fare orders with the Electronic Miscellaneous Document (EMD) system [62]. Consequently, it is proposed that the implementation of electronic ticketing with EMD will revolutionize the experience for passengers, airlines, and travel agencies and will propel the total digitalization process in the industry (Figure 2) [60].

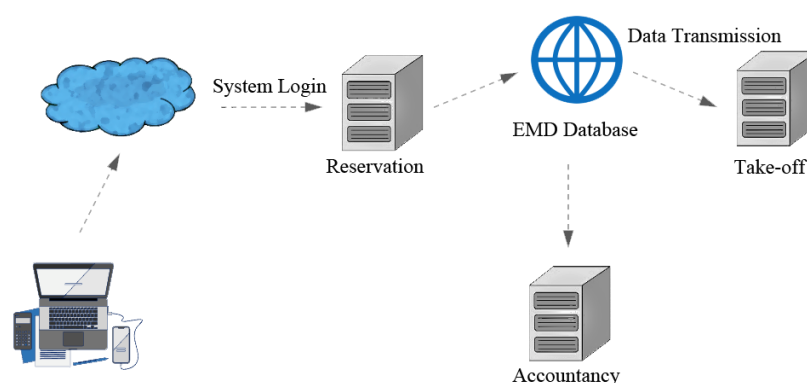


Figure 2. Airline EMD processing system.

Numerous successful examples of online product sales are linked to categories such as travel, entertainment, books, music, and computers [63]. As reported by the International Air Transport Association, the global Revenue Passenger Kilometers (RPK) growth has surged from USD 5.3 billion to USD 7.4 billion since 2012. This signifies a growth rate of 30% over 5 years. Consequently, in order for airlines to expand, they need to embrace e-ticketing, a strategy that facilitates more efficient passenger reach and ongoing cost reduction. According to Backes et al. [64], since June 2008, the airline industry has made a 100% transition to electronic ticketing. López-Bonilla and López-Bonilla [63] believe that e-ticketing can be considered one of the biggest innovations in the aviation industry. The implementation of e-ticketing by airlines has led to numerous benefits, including improved record keeping, financial control, reduced time-consuming factors, decreased reliance on indirect sales (via travel agents), and enhanced customer convenience through self-service options. All of these benefits have eliminated costs, resulting in annual savings of USD 3 billion. The savings are due to the fact that the cost of the e-ticketing system is USD 1 per ticket compared to the price of a paper ticket, which is USD 10 per ticket [64].

3.3. Paperless Air Traffic Control Tower

On the airport surface, each line or “stripe” represents an aircraft or other form of traffic. The paper strips serve as a symbolic representation of the flight or traffic data that is accessible within the centralized Air Traffic Control (ATC) system [65,66]. Once these strips are printed, air traffic controllers handle the evolving traffic situation by annotating these paper strips. The aforementioned handwritten notes possess significant operational information that is pertinent not only to the controller in question, but also to other controllers situated within or outside the tower [67]. The process of converting the strips into a digital format enables controllers to input instructions into the central Air Traffic Control (ATC) system, thereby facilitating access to this information by a larger number of controllers and other relevant stakeholders, including airport personnel and airlines [65]. Furthermore, it should be noted that paper strips are positioned within holders that necessitate additional manual manipulation, thereby contributing to the generation of a disruptive auditory atmosphere within the tower [68]. Bos et al. [69] presented the prototyping of an Electronic Flight Strip (EFS) system user interface for the air traffic control tower at Amsterdam Airport Schiphol and the evaluation of the human part in a loop in the National Aeronautics and Space Laboratory (NLR) tower research simulator as a realistic traffic example. It is argued that prototyping and human-in-the-loop assessment are necessary to obtain system and application requirements for an EFS system from ATC in the Netherlands. Within ATC towers, the implementation of EFS systems is currently underway to replace traditional paper flight strips [70,71]. The following items are the basis for flight strips [72]:

1. Convey flight information to the controllers.
2. Enable the controller to execute their directives.
3. Develop and sustain a cognitive representation of the aircraft in one’s mind while exercising control over it.
4. Enable the effortless transfer of flight control responsibilities between multiple controllers.

3.4. Paperless Cockpit

Since the dawn of aviation, pilots have consistently emphasized the importance of paper-based information. This includes various forms of data, such as navigation charts during the early stages of aviation, as well as extensive collections of aircraft performance data, checklists, weather advisories, and other relevant materials that they presently carry [73]. The development of the paperless cockpit concept has been facilitated by the accessibility of electronic data [74]. Furthermore, the inclusion of an independent display within the aircraft’s cockpit, separate from the primary flight instruments, facilitates the presentation of supplementary information. This technology not only grants users the ability to access information that was previously stored in physical flight bags, but it also offers real-time updates on weather conditions, airport mapping for the prevention of runway incursions, and other data that are required by the Federal Aviation Administration (FAA). The development of the Integrated Crew Information System (ICIS) by Avionitek aims to reduce crew workload and facilitate the transition towards a paperless cockpit (Figure 3) [73].

The prevalence of paper-based materials within the aircraft cockpit has resulted in an augmented burden on both pilots and co-pilots, particularly during the most crucial phases of flight. The current state of imaging and related computer technology shows that data can be quickly and easily retrieved in a digital environment where there are no physical documents. The emergence of new systems—designed to enhance flight safety as mandated by the FAA—coupled with advancements in display technology, lays the groundwork for the realization of a paperless cockpit [73,75].



Figure 3. ICIS indicator in the aircraft environment [73].

3.5. Aircraft Maintenance–Repair

Aircraft operations and maintenance need to be carefully planned to ensure safety [76]. The obligatory requirement necessitates a substantial volume of documentation as evidence of proper procedural adherence. The documentation undergoes a series of stages involving various parties, such as Original Equipment Manufacturers (OEMs), leaseholders, Maintenance Repair and Overhaul Organizations (MROs), suppliers, and airlines [5–7]. Each individual must effectively oversee their transactions and data in accordance with the regulations set forth by the registration authority. Digitalization is gradually modernizing the conventional procedures of aircraft operations and maintenance [9]. Although the adoption of key enabling solutions and initiatives, such as digital aircraft procedures (commonly referred to as paperless aircraft), operations and maintenance, e-signatures, and Radio Frequency Identification (RFID), brings about notable benefits, it also introduces supplementary obstacles [77,78]. The primary goal of these implementations is to harmonize the flow of information about technological solutions, which can be attained by adopting universally accepted requirements and unified information standards. IATA has developed an implementation guide for paperless aircraft operations with the aim of providing support to its partners and researchers [10].

Paper-based maintenance record keeping still dominates the aviation industry [79]. However, digital formats are increasingly being adopted, with aviation organizations such as the International Civil Aviation Organization (ICAO) [80] and IATA launching Guidelines for the Adoption of Electronic Aircraft Maintenance Records (EAMR) [61], Guidelines for Admission [81], and the Paperless Aircraft Guide [82]. The guidelines and materials set out requirements and standards to be considered by Civil Aviation Authorities (CAAs) and stakeholders [83]. Digitalization intrinsically brings significant benefits for stakeholders, such as increased efficiency, time saving, and cost reduction. However, there are also practical aspects that need to be addressed. These include the establishment of common regulatory standards, acceptance by stakeholders, and adoption of universal requirements (e.g., lease agreements and management), as well as the harmonization of technological solutions for efficient access and data interoperability [10].

One of the major challenges for DAOM is the establishment of common regulatory standards, as each regulatory body applies different standards (Figure 4) [78,84]. The absence of common regulatory standards has emerged as a significant barrier to achieving comprehensive digitalization globally. This is mainly due to the lack of clarity on the position of all regulatory bodies on digital care records. It is clear that by resolving this issue, stakeholders will be more willing to take advantage of the opportunities of the new approach and thereby increase its global acceptance. A prime example in this regard is e-signature technology [85]. Although e-signature technology is crucial for the digital transformation of the aviation industry, it has not been widely accepted until recently.

Establishing a universal regulatory standard would bridge critical gaps in areas such as security, data access, and interoperability. Stakeholder acceptance is intimately tied to the establishment of common standards. It is not practical to accept an approach or process that could potentially be rejected by one or more regulatory authorities. Therefore, the adoption of digital systems for aircraft operation and maintenance and the planning, documentation, and delivery of documentation for the aircraft leasing process play an important role. The report on Digital Aircraft and Engine Lease Returns [86] outlines the challenges associated with delivering and returning aircraft and enumerates the benefits of digital maintenance record keeping like improved accessibility and facilitating aircraft transition.

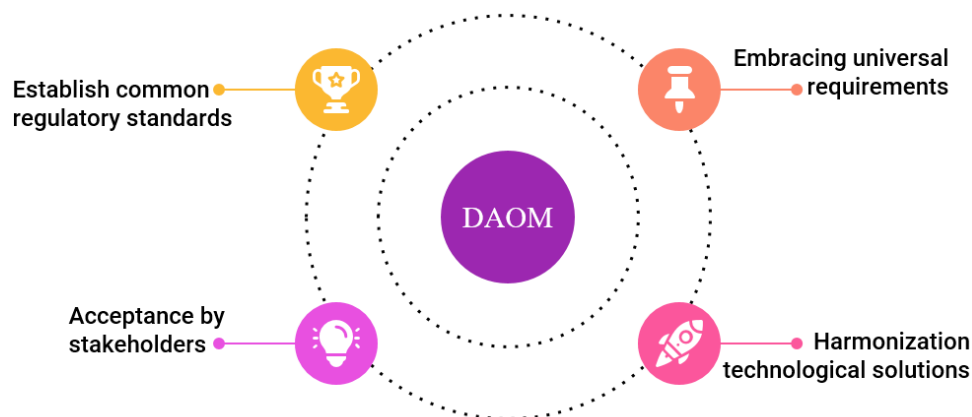


Figure 4. DAOM layout.

Harmonizing digital solutions is only feasible when the digital systems utilized by stakeholders conform to identical or comparable standards. This alignment is crucial to ensuring data interoperability and optimal utilization of underlying technologies and systems. For example, when an aircraft is leased to more than one airline operator and then receives maintenance work from more than one MRO over a period of time, the collection of aircraft records eventually becomes complex [78]. Digital records management systems are being implemented in aviation to digitize past maintenance records and manage them through existing systems on the market such as AerData STREAM, AMOS, and EmpowerMx, implementing fully electronic task cards with electronic signatures [64]. Given the aviation industry's ongoing struggle to transition to a paperless system, only a fraction of airlines has adopted digital records. The aviation industry presents a promising domain for substantial enhancements that can yield numerous favorable outcomes across the entire value chain [87]. Research on digital transformation in the aviation industry is categorized under three important headings: return delivery transition, digital records, and maintenance records.

According to estimations provided by IATA and Boeing, it is projected that approximately 50% of the global aircraft fleet will be subject to operating leases by the year 2020. At the IATA Paperless Aircraft Operations Conference 2016, Doug Walker, Vice President of Technical at AerCap, stated that electronic records offer a greatly improved and searchable level. The challenges associated with implementing a paperless strategy can be divided into three areas [88]:

1. Authorities: There is a requirement for supplementary regulatory guidance material that extends beyond the Federal Aviation Administration (FAA) and is universally recognized.
2. Leaseholders: It is necessary to advocate for the implementation of a standardized framework for electronic records.
3. Airlines: The concept can be categorized as a fear of novelty or a preference for familiarity.

During the 2016 AIRCRAFT IT MRO Conference, Tim Scott, Vice President of Technical Services at AVITAS, suggested that the implementation of digital or electronic records could mitigate many of the previously mentioned issues. Scott further claimed that digital records streamline the process of sending the bulk of the package to leaseholders, thereby reducing the time required for on-site record review [89]. Moreover, a white paper presented at the AIRCRAFT IT MRO Conference [90], titled “Digital aircraft and engine lease returns”, explained the role of digital records in facilitating aircraft transition. The paper emphasized the need for accurate and efficient handling of all maintenance records for owned or leased aircraft. Consequently, it was suggested that well-organized digital records could decrease delays in aircraft transitions, leading to substantial savings for airlines and leaseholders [64].

Aircraft maintenance record keeping is a critical procedure for maintaining the airworthiness of an aircraft. At present, the storage of aircraft maintenance logbooks predominantly relies on physical logbooks that are situated on the aircraft itself or held by the owner [91]. The primary risk inherent in the practice of maintaining a physical logbook is the possibility of its loss or theft. In the absence of a logbook documenting proper maintenance, an aircraft, regardless of its impeccable condition, cannot be deemed airworthy [92]. In the report titled “The Relationship Between Aircraft Value and Maintenance Condition”, Mr. Shannon Ackert, Senior Vice President of Commercial at Jackson Square Aviation and owner of Aircraft Monitor, asserts that the maintenance condition of an aircraft significantly influences its value. He contended that inconsistencies, errors, and missing documentation in maintenance records often reflect their poor quality and can lead to substantial added costs for the owner or operator [64].

Transitioning to paperless operations in maintenance and engineering offers several value propositions, including:

- Optimized integration between engineering, maintenance, and operations;
- Real-time tracking and reporting of aircraft status;
- The provision of real-time aircraft performance visibility;
- Real-time record keeping;
- Real-time monitoring of airworthiness compliance;
- Visibility of maintenance tasks and work order progress according to plan;
- Increased data accuracy and completeness support informed decision making [82].

As a key driver of economic and technical growth in the aviation industry, aircraft loan financing necessitates adherence to traditional loan transaction principles, especially when the government or public financing is involved [93]. An aircraft is a tangible object with a specific function and high resale value; therefore, it can be used as collateral for the fulfillment of a debt payment obligation [94]. In 2016, consulting firm Oliver Wyman highlighted in its report, “The Future of Technology in Mortgage Lending”, that a breakthrough seems imminent due to new technologies, the rise of Financial Services (Fintech), and a competitive origination market. According to the report, mortgage applications will become paperless, and underwriting will become an increasingly automated and digital system [64].

3.6. Paperless Aircraft Operations within IATA

IATA has introduced the Paperless Aircraft Operations in the Technical Operations program. This initiative, which draws inspiration from the “Simplifying Business” program, aims to support airlines in identifying opportunities and implementing solutions to enhance the efficiency of aircraft operations in various domains, including technical operations [95]. Operations encompass a range of activities within the aviation industry, such as the maintenance of aircraft, the management of parts supply chains, the coordination of logistics, and the transfer of aircraft assets. The goal is to leverage IATA’s unique position to encourage industry-wide adoption of modern technologies and processes that meet defined standards. By adopting paperless operations, industry stakeholders aim to enhance efficiency and effectiveness, contributing to the industry at large through engineering and maintenance across the aircraft lifecycle.

Similarly, incorporating diverse paperless technologies in operational aspects will enable the optimization of airline processes, both internally and at the interfaces with stakeholders [96]. However, it is important to note that implementing paperless strategies in airline operational areas presents considerably more challenges compared to e-ticketing [82]. The global e-ticketing rate reached 96.5% [97] by the 1 June 2008 deadline that IATA had originally set, and smaller, less developed airlines were finding it difficult to deal with worries about information technology automation and the exchange of ticketing information for routes between airlines.

3.7. The Research Gap

Considering the relationship between aviation operations and paper (green airports, electronic ticketing, paperless cockpit and air traffic control tower, and paperless aircraft operations within IATA), which is examined in detail in Section 3, the literature gap that underpins the motivation for this research is as follows:

While the literature extensively covers a range of facets pertaining to paperless practices within the aviation industry, including electronic ticketing, air traffic control procedures, cockpit operations, aircraft maintenance, and initiatives introduced by the IATA, an essential void becomes apparent. The literature effectively delineates the theoretical framework, offering insights into potential benefits, projected rates of global e-ticketing adoption, and potential cost savings associated with paperless practices. However, this body of literature appears to be somewhat bereft of actual, empirical data that would substantiate these claims. Thus, the research presented here stands as a critical response to this discernible gap in the literature. It endeavors to transcend the theoretical boundaries of prior studies by engaging with the tangible, real-world operations of a business entity operating within the aviation sector. The research is uniquely poised to offer a granular examination of the tangible savings and environmental implications brought by the adoption of paperless practices within this specific aviation context.

4. Paper Cost in Aircraft Maintenance and Repair Operations: An Evaluation of a Maintenance Organization with Real Data

This study uses the first independent maintenance center in Turkey, which operates at the world's first lean-built Istanbul Sabiha Gökçen International Airport (SAW). Situated strategically between Europe and Asia, the maintenance center serves as an ideal partner for airlines not only in Turkey, but also in Europe, Russia, the Middle East, and Africa. The construction of the hangar encompassed a land area measuring 60,000 square meters. The architectural design of the building featured a contemporary three-story structure incorporating a hangar space spanning 15,400 square meters. The establishment additionally encompasses a total area of 24,800 square meters, which is allocated for workshop, office, and warehouse purposes. Furthermore, the facility includes a spacious engine workshop spanning an area of 6000 square meters, along with the prestigious European Union Aviation Safety Agency (EASA) (Part 145) authorization. Designed according to lean management principles, it provides timely and efficient services ranging from airframe, engine, and component maintenance and repair to aircraft painting for wide and narrow-body aircraft. The cost implications of paper usage are evaluated in the context of planning aircraft maintenance and repair operations. The maintenance organization in question employs a total of 850 individuals and undertakes maintenance, repair, and overhauls on approximately 15 aircrafts each month.

Each organization's ongoing business plan should identify relevant categories and provide forecasts and return on investment estimates grounded in as much objective data as possible. For instance, improvements in airworthiness achieved via paperless operations can translate to enhanced safety and compliance, thereby contributing non-financial benefits to the business plan. However, the improvements can also lead to reductions in rework and other efficiencies that are directly reflected in the bottom line [87]. IATA, which represents 83% of airline operators, has estimated that the percentage of aircraft in leasing activities

will increase from 5% in 1980 to nearly 50% by 2020 [82]. Nonetheless, in a white paper on redelivery, the International Bureau of Aeronautics (IBA) reported that “Records” accounted for 60% of the challenges in on-time and on-budget redelivery, with engines, airframe, and interior means accounting for the rest [98]. As records of maintenance status and required processes are a large part of the value of an aircraft or component, it is critical that digital systems complement the process through data integration, retrieval, and analytics [99]. The value proposition arising from paperless operations should eclipse any increase in costs. Where relevant, it is advisable to incorporate notes to this effect in corresponding sections of business plans and annual budget structures. Table 1 [10] illustrates considerations in areas where costs may potentially increase.

Table 1. Cost of paperless operations.

Areas of Cost Expansion	Reasons
Data transmission costs	Paperless transactions involve the transmission of data electronically through various means, such as satellite, Wi-Fi, and telecommunication networks like 4G.
	As the volume of data increases, there is a likelihood of concurrent increases in the expenses related to the transportation of data, which necessitates quantification and inclusion in business plans.
Infrastructure	Investment in infrastructure may be needed to support PAO:TO.

Aircraft undergo routine and non-routine maintenance, including repair, retrofitting, inspection, and modification, to ensure safe flights [100]. Airworthiness is largely maintained through aircraft maintenance activities [101]. Aircraft maintenance encompasses a range of activities aimed at restoring an item to a condition that allows it to be used effectively. These activities include servicing, repair, modification, overhaul, inspection, and condition determination [102]. The primary objective of aircraft maintenance is to guarantee the availability of a fully operational aircraft for an airline while minimizing expenses. Maintenance costs make a significant contribution to the cost of ownership of an aircraft. Maintenance costs typically account for 10–20% of the operating costs associated with the aircraft [103]. Maintenance planning involves the printing of hard copy task cards for all necessary operations, including maintenance, repair, retrofitting, inspection, and modification operations, non-routine cards for findings discovered following the necessary inspection and control operations, and customer additional request cards while performing aircraft maintenance activities, which are one of the most crucial operations after airworthiness [101]. Accordingly, Figure 5 shows the cost graph of the black papers spent until August 2021/2022 by an aircraft maintenance, repair, and overhaul organization operating in Istanbul Sabiha Gökçen Enterprise.

Analysis of the paper consumption graph for 2021 in Figure 5 reveals that a total of 1,066,827 sheets of black paper were used from January to August. Of these, the highest consumption was 205,348 (19%) in August, and the lowest consumption was 97,082 (9%) in March. On the other hand, the paper consumption graph for 2021 in Figure 6 shows that a total of 133,912 pieces of colored paper were used between January and August, with the highest consumption of 22,992 (17%) in August and the lowest consumption of 12,020 (9%) in February. In addition, looking at the paper consumption graph for 2022, it is seen that a total of 1,119,243 pieces of black paper (Figure 7) and 147,064 pieces of colored paper (Figure 8) were used in January–August.

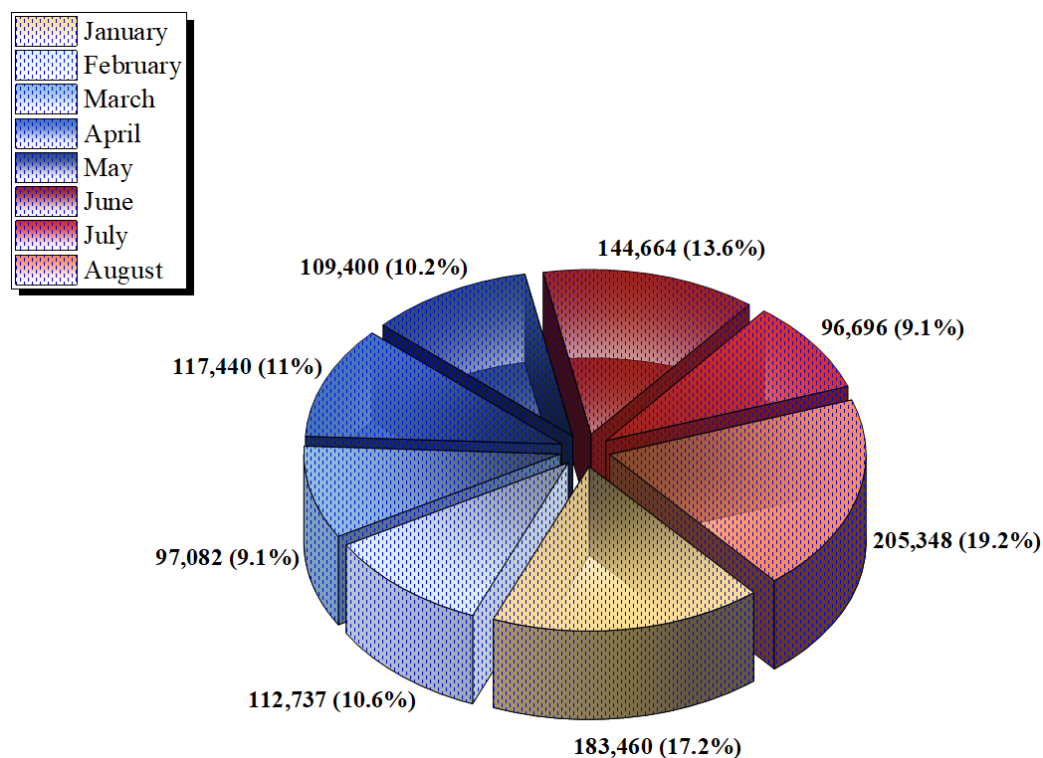


Figure 5. The 2021 cost of black paper spent (USD) for 8 months.

Considering the consumption of black and colored paper, the highest consumption was 157,753 (14%) in February and 24,514 (17%) in July, respectively (Figures 7 and 8). The lowest consumption of both black and colored paper was 111,538 (10%) and 15,009 (10%) in June, respectively. Within the scope of the relevant data, it is seen that the rates decrease or increase according to months and years. It is stated that the reason for this is that, due to the increase in tourism flights between May and October, operator companies prefer to have their maintenance performed between May and October. Therefore, seasonal paper usage increases proportionally during this period (Figures 9 and 10). In addition, paper operations create a need for personnel due to the increase in workload compared to personnel employment. Since an increase in personnel means an increase in output, there is an increase in paper in direct proportion. Among the costs spent on paper, hardware and printing account for the highest usage. When paper usage areas were considered, black paper was used in maintenance manuals, while colored paper was used in documents with tracking cards and yellow-red-green warning notifications. Table 2 presents monthly average data on paper usage broken down by department.

Table 2. Paper consumption by departments.

Departments	Black Paper Consumption (pcs)	Colored Paper Consumption (pcs)
Component Shop	13,131	2315
Production Planning	79,932	13,770
Base Maintenance	54,716	2245
Engineer	19,724	5381

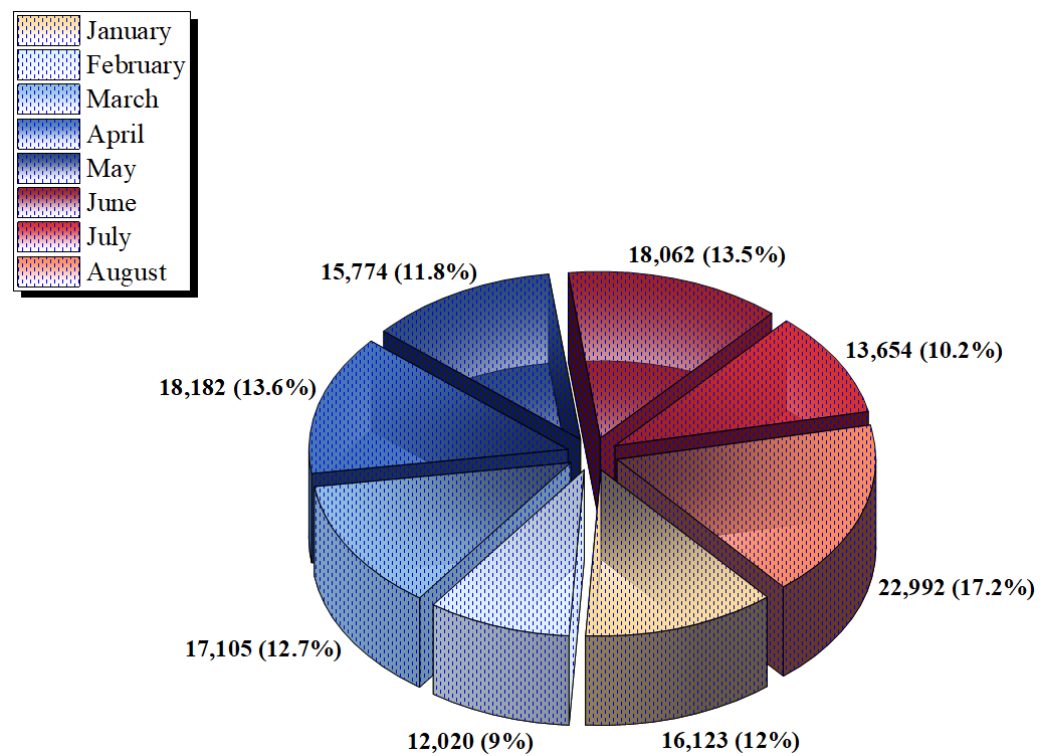


Figure 6. The 2021 cost of colorful paper spent (USD) for 8 months.

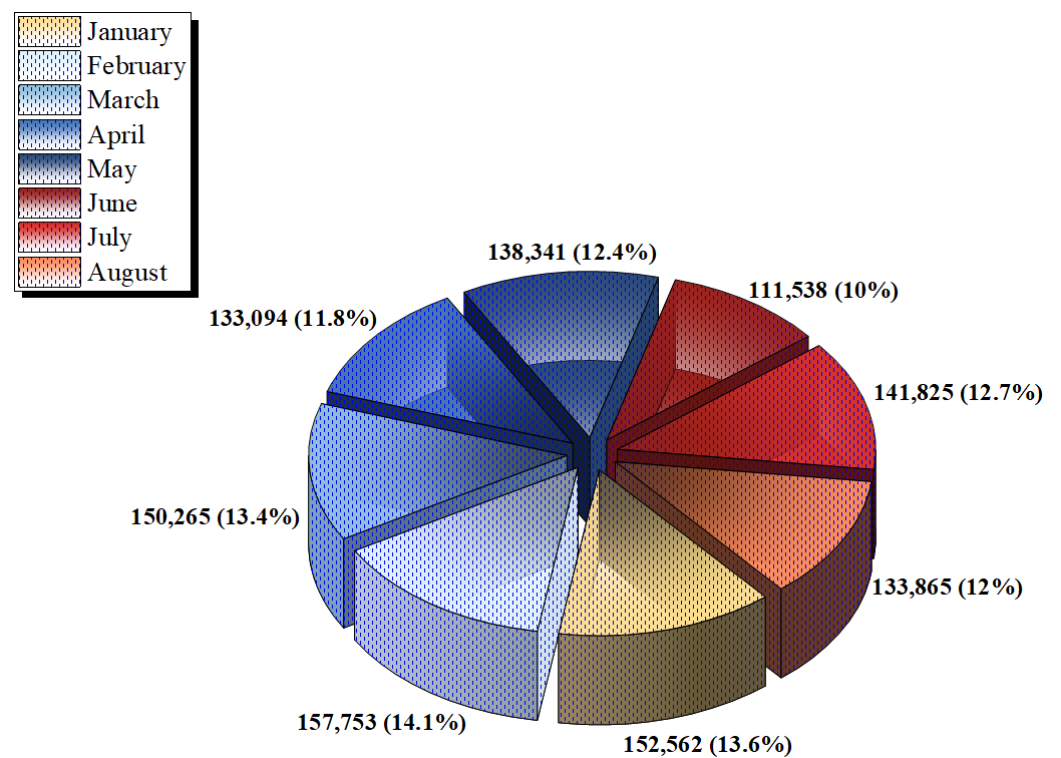


Figure 7. The 2022 cost of black paper spent (USD) for 8 months.

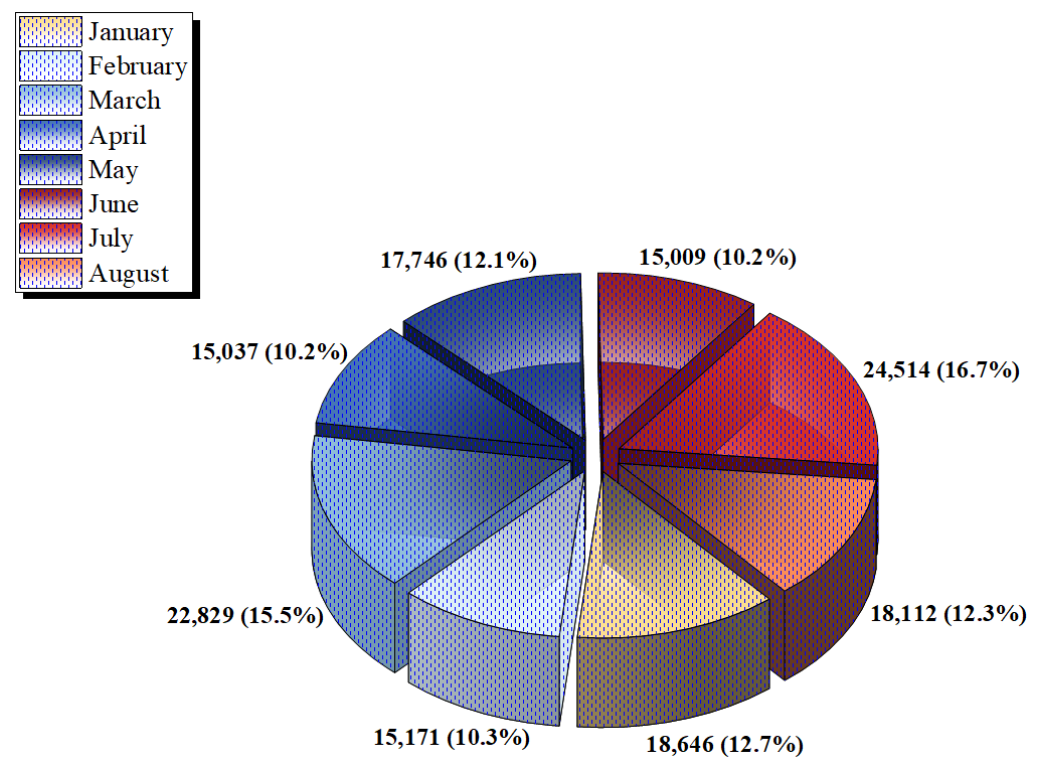


Figure 8. The 2022 cost of colorful paper spent (USD) for 8 months.

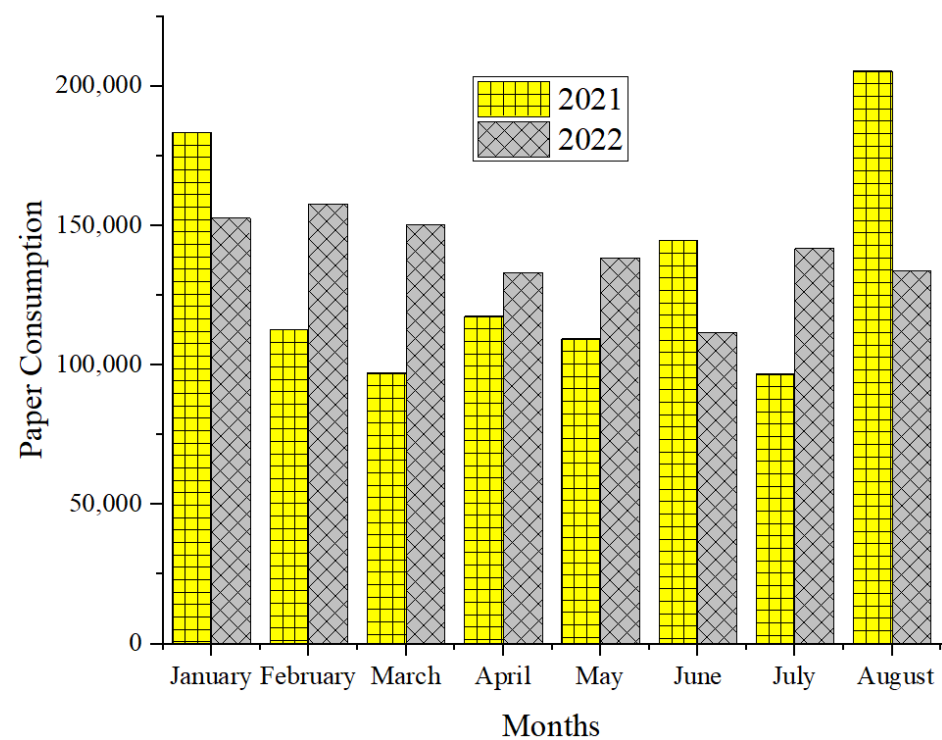


Figure 9. Comparative representation: black paper spent graph for 8 months.

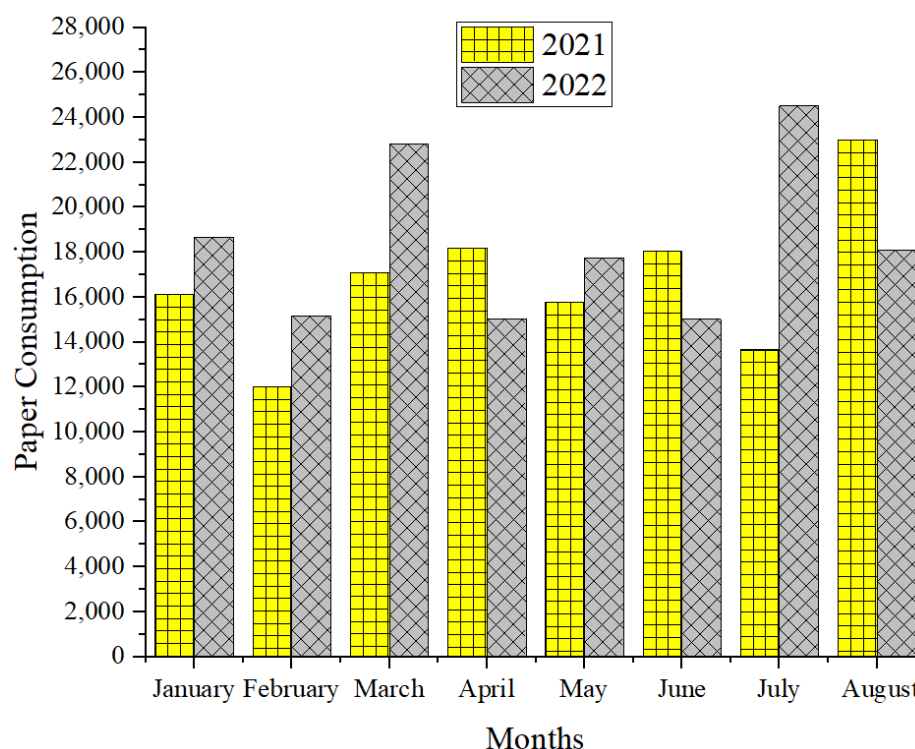


Figure 10. Comparative representation: colorful paper spent graph for 8 months.

5. Challenges and Considerations

Paperless aircraft maintenance practices present a promising path for the aviation industry. While challenges such as regulatory compliance, data security, and technological integration exist, the benefits in terms of cost savings, environmental sustainability, and operational efficiency are substantial. The challenges and contemplations inherent in the process of transitioning to paperless aircraft maintenance can be categorized as follows:

Regulatory requirements: An eminent challenge associated with the adoption of paperless aircraft maintenance practices pertains to the intricate navigation of rigorous regulatory stipulations inherent to the aviation industry. Globally recognized aviation authorities assert compelling mandates necessitating the retention of comprehensive records encompassing aircraft maintenance, repair, and operational data in a tangible, printed format for a stipulated duration. This regulatory framework imposes a noteworthy constraint, effectively rendering aircraft as carriers of their own extensive documentation burden.

Data security: Within the aviation sector, the preservation of the security and confidentiality of sensitive maintenance data assumes paramount importance. The transition towards paperless practices mandates the establishment of a robust digital infrastructure that can effectively uphold the confidentiality and integrity of pivotal maintenance records. Integral to this infrastructure must be cybersecurity measures, meticulously devised and diligently enforced, to counteract the looming threats of data breaches and unauthorized access, thus fortifying the protective shield surrounding critical maintenance information.

Competitive landscape: Operating within a fiercely competitive milieu, the aviation industry is emblematic of a sector that thrives on cutting-edge technological advancements. The efficacious adoption of paperless maintenance practices necessitates the adept integration of digital solutions into the existing operational framework of airline companies. This integration mandate encompasses the provisioning of comprehensive training for personnel and the assurance of the reliability and user friendliness of digital platforms. In an industry where precision and efficiency are crucial, these considerations emphasize the imperative of a seamless transition to paperless maintenance procedures.

Integration of paperless solutions with legacy systems: The amalgamation of paperless solutions with pre-existing legacy systems constitutes a complex undertaking, necessitating

meticulous planning and implementation. This complex process highlights the importance of strategic deliberation and foresight to seamlessly bridge the technological divide and ensure the harmonious co-existence of modern digital solutions with entrenched legacy infrastructure.

Skill of maintenance personnel in digital proficiency: Achieving a seamless transition in the adoption of paperless maintenance practices hinges fundamentally upon the proficiency of maintenance personnel in utilizing digital tools and acclimating themselves to novel workflows. This critical element is indispensable to ensure the harmonious integration of digital technologies into the operational framework, thereby upholding the operational efficiency and effectiveness of the transition process.

6. Conclusions

As our world's natural resources dwindle, we confront escalating environmental issues. The first of the two most dangerous environmental problems is the increasing greenhouse effect, and the second is the global change in climate due to the depletion of the stratospheric ozone layer. Despite aviation being the least environmentally damaging among the transportation sectors, the emergence of global dangers necessitates new, less harmful solutions. These solutions aim to minimize errors and risks, expedite communication, and enhance safety and sustainability. Given the critical role of paperless aircraft maintenance activities in maintaining airworthiness, the documentation of maintenance is as significant as the work itself. Paperless maintenance activities have fostered the development of green airport projects spanning air traffic, cockpits, and ticketing, further promoting paperless strategies and digitalization. Thus, it has had an impact on many issues, such as data access, data processing, digitalization, and software development, such as Industry 3.0 and 4.0 applications, which have contributed to major developments in the aviation industry. Paperless applications in aviation operations conserve resources and streamline every stage of maintenance, foster new software creation, and facilitate the transition from paper-based to AI-supported processes. Therefore, developing technology will provide convenience at every stage of our lives. Therefore, by replacing paper with digitalization in aviation operations, efficiency in aviation will increase, time spent on paper will be saved, and man hours will be saved. In addition, the cost of paper applications, the space required for the storage of paper, and errors are minimized with paperless applications and simultaneous follow ups.

The aviation industry is characterized by the utilization of cutting-edge technological tools and a highly competitive environment. In light of technological advancements, airline companies are seeking to incorporate these developments into their operations. Consequently, their objective is to enhance their operational efficiency through the more precise and effective utilization of their resources. A private aircraft maintenance, repair, and overhaul organization in Turkey contributed to the completion of this study. In recent years, a large amount of emphasis has been placed on digitalization (paperlessness) to reduce costs, store data securely, and contribute to a greener environment. Costs for both color and black paper used for maintenance records in 2021/2022 were considered. The use of 2,000,000 sheets of paper costs approximately USD 21,000, representing the use of 500 trees. In addition, 11,200 gallons of oil, 90 cubic meters of storage space, and 119,000 kW of energy will be used.

The company has embraced digitalization in line with technological developments and has implemented various projects to enhance paperless maintenance practices. Accordingly, instead of printing maintenance manuals, 20 tablets were distributed to the personnel, and access to Boeing Toolbox and Airbus Airnav was provided via tablets. This enabled electronic access to maintenance documents, significantly reducing the volume of printed materials. In consideration of the aforementioned data, the company's transition to paperless maintenance has resulted in savings of approximately 168,000 pages of black print and 24,000 pages of color print.

The limitations and future works of this research are as follows:

- The study primarily focuses on a single maintenance, repair, and overhaul organization located at Istanbul Sabiha Gökçen International Airport. While this case study provides valuable insights, it limits the generalizability of the findings to the broader aviation industry. A more extensive and diverse sample of aviation organizations would provide a more comprehensive perspective.
- The aviation industry is highly regulated, and the research does not delve into the challenges or potential compliance issues associated with transitioning to paperless maintenance records. Addressing these regulatory concerns would provide a more comprehensive analysis.
- While the work mentions using tablets and digital tools, it does not delve into the potential technological challenges or barriers that organizations may encounter when implementing paperless systems. Investigating these challenges would provide a more holistic view of the transition process.
- Given the sensitive nature of aviation maintenance records, the study lacks a thorough investigation of data security measures and considerations when adopting paperless practices. An exploration of data encryption, access controls, and data breach risks would be beneficial.
- The work does not incorporate the perspectives of key stakeholders, such as aviation maintenance personnel, regulatory authorities, or passengers. Including these viewpoints through surveys or interviews would offer a more comprehensive understanding of the implications of transitioning to paperless practices.

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References

1. Vafaenezhad, T.; Tavakkoli-Moghaddam, R.; Cheikhrouhou, N. Multi-objective mathematical modeling for sustainable supply chain management in the paper industry. *Comput. Ind. Eng.* **2019**, *135*, 1092–1102. [\[CrossRef\]](#)
2. Tian, L.; Zhang, X.; Shao, Y.; Qin, W. The Impact of the Corsia on CO₂ Emissions. *SSRN J.* **2022**. [\[CrossRef\]](#)
3. Jiang, X.; Zhang, Y.; Li, Y.; Zhang, B. Forecast and analysis of aircraft passenger satisfaction based on RF-RFE-LR model. *Sci. Rep.* **2022**, *12*, 11174. [\[CrossRef\]](#) [\[PubMed\]](#)
4. Henderson, I.L. Aviation safety regulations for unmanned aircraft operations: Perspectives from users. *Transp. Policy* **2022**, *125*, 192–206. [\[CrossRef\]](#)
5. Goncalves, C.D.; Kokkolaras, M. Modeling the relationship between aviation original equipment manufacturers and maintenance, repair and overhaul enterprises from a product-service system perspective. In Proceedings of the 21st International Conference on Engineering Design (ICED 17) Vol 3: Product, Services and Systems Design, Vancouver, BC, Canada, 21–25 August 2017.
6. Francis, G.; Humphreys, I.; Ison, S. Airports' perspectives on the growth of low-cost airlines and the remodeling of the airport–airline relationship. *Tour. Manag.* **2004**, *25*, 507–514. [\[CrossRef\]](#)
7. Efthymiou, M.; McCarthy, K.; Markou, C.; O'Connell, J.F. An Exploratory Research on Blockchain in Aviation: The Case of Maintenance, Repair and Overhaul (MRO) Organizations. *Sustainability* **2022**, *14*, 2643. [\[CrossRef\]](#)
8. Rodrigues, D.; Lavorato, P. Maintenance, Repair and Overhaul (MRO) Fundamentals and Strategies: An Aeronautical Industry Overview. *Int. J. Comput. Appl.* **2016**, *135*, 21–29. [\[CrossRef\]](#)
9. Novák, A.; Sedláčková, A.N.; Bugaj, M.; Kandera, B.; Lusiak, T. Use of Unmanned Aerial Vehicles in Aircraft Maintenance. *Transp. Res. Procedia* **2020**, *51*, 160–170. [\[CrossRef\]](#)
10. Abdallah, A.A.; Fan, I.-S. Emerging Challenges of Digital Aircraft Operations and Maintenance: A Knowledge Management Perspective. *SSRN J.* **2020**. [\[CrossRef\]](#)

11. Sheehan, W.M. International Air Transport Association (IATA). *J. Air L. Com.* **1953**, *20*, 206.
12. Turner, B. International Air Transport Association (IATA). In *The Statesman's Yearbook*; Turner, B., Ed.; Palgrave Macmillan UK: London, UK, 2010; pp. 45–46. ISBN 978-0-230-20603-8.
13. Wu, H.; Liu, Y.; Ding, Y.; Liu, J. Methods to reduce direct maintenance costs for commercial aircraft. *Aircr. Eng. Aerosp. Technol.* **2004**, *76*, 15–18. [\[CrossRef\]](#)
14. Kilpi, J.; Töyli, J.; Vepsäläinen, A. Cooperative strategies for the availability service of repairable aircraft components. *Int. J. Prod. Econ.* **2009**, *117*, 360–370. [\[CrossRef\]](#)
15. Tretheway, M.W.; Markhvida, K. The aviation value chain: Economic returns and policy issues. *J. Air Transp. Manag.* **2014**, *41*, 3–16. [\[CrossRef\]](#)
16. Yu, D.; Michael, T. The Ukrainian Conflict: Impact on the Aviation Finance and Leasing Industry. *J. Struct. Financ.* **2022**, *28*, 33–41. [\[CrossRef\]](#)
17. Merrild, H.; Damgaard, A.; Christensen, T.H. Recycling of paper: Accounting of greenhouse gases and global warming contributions. *Waste Manag. Res.* **2009**, *27*, 746–753. [\[CrossRef\]](#)
18. Tiwari, U.; Diwan, T.D.; Sahu, N. Save green, go green through paperless. *Int. J. Recent Trends Eng. Res. (IJRTER)* **2016**, *3*.
19. WWF. *The Forest Industry in the 21st Century*; WWF: Godalming, UK, 2001.
20. Sinclair, P.; Walton, J. Environmental reporting within the forest and paper industry. *Bus. Strat. Environ.* **2003**, *12*, 326–337. [\[CrossRef\]](#)
21. Wang, J. Creating a Paperless Classroom with the Best of Two Worlds. *J. Instr. Pedagog.* **2010**, *2*, 1–22.
22. Hwang, G.; Lee, J.; Park, J.; Chang, T.-W. Developing performance measurement system for Internet of Things and smart factory environment. *Int. J. Prod. Res.* **2017**, *55*, 2590–2602. [\[CrossRef\]](#)
23. Shariatzadeh, N.; Lundholm, T.; Lindberg, L.; Sivard, G. Integration of Digital Factory with Smart Factory Based on Internet of Things. *Procedia CIRP* **2016**, *50*, 512–517. [\[CrossRef\]](#)
24. Chen, J.; Yang, J.; Zhou, H.; Xiang, H.; Zhu, Z.; Li, Y.; Lee, C.-H.; Xu, G. CPS Modeling of CNC Machine Tool Work Processes Using an Instruction-Domain Based Approach. *Engineering* **2015**, *1*, 247–260. [\[CrossRef\]](#)
25. Zakoldaev, D.A.; Shukalov, A.V.; Zharinov, I.O.; Zharinov, O.O. Modernization stages of the Industry 3.0 company and projection route for the Industry 4.0 virtual factory. *IOP Conf. Ser. Mater. Sci. Eng.* **2019**, *537*, 32005. [\[CrossRef\]](#)
26. Ahmad, R.; Kamaruddin, S. An overview of time-based and condition-based maintenance in industrial application. *Comput. Ind. Eng.* **2012**, *63*, 135–149. [\[CrossRef\]](#)
27. Fritzsche, R.; Lasch, R. An Integrated Logistics Model of Spare Parts Maintenance Planning within the Aviation Industry. *World Acad. Sci. Eng. Technol.* **2012**. [\[CrossRef\]](#)
28. Filipov, V.; Vasilev, P. *Manufacturing Operations Management—The Smart Backbone of Industry 4.0*; Springer: Cham, Switzerland, 2016; p. 19.
29. Savage, D. Has MES Reached Maturity in the Pharmaceutical Industry? ATS Global: Rochester Hills, MI, USA, 2018.
30. Vinhais, J. MES Reduces FDA Compliance Costs. *Quality* **2004**, *43*, 38.
31. Hwang, Y.-D. The practices of integrating manufacturing execution systems and Six Sigma methodology. *Int. J. Adv. Manuf. Technol.* **2006**, *31*, 145–154. [\[CrossRef\]](#)
32. Saenz de Ugarte, B.; Artiba, A.; Pellerin, R. Manufacturing execution system—A literature review. *Prod. Plan. Control* **2009**, *20*, 525–539. [\[CrossRef\]](#)
33. Reinhardt, I.C.; Oliveira, J.C.; Ring, D.T. Current Perspectives on the Development of Industry 4.0 in the Pharmaceutical Sector. *J. Ind. Inf. Integr.* **2020**, *18*, 100131. [\[CrossRef\]](#)
34. Oliva, F.L.; Teberga, P.M.F.; Testi, L.I.O.; Kotabe, M.; Del Giudice, M.; Kelle, P.; Cunha, M.P. Risks and critical success factors in the internationalization of born global startups of industry 4.0: A social, environmental, economic, and institutional analysis. *Technol. Forecast. Soc. Chang.* **2022**, *175*, 121346. [\[CrossRef\]](#)
35. Helu, M.; Hedberg, T. Enabling Smart Manufacturing Research and Development using a Product Lifecycle Test Bed. *Procedia Manuf.* **2015**, *1*, 86–97. [\[CrossRef\]](#)
36. Zheng, P.; Wang, H.; Sang, Z.; Zhong, R.Y.; Liu, Y.; Liu, C.; Mubarak, K.; Yu, S.; Xu, X. Smart manufacturing systems for Industry 4.0: Conceptual framework, scenarios, and future perspectives. *Front. Mech. Eng.* **2018**, *13*, 137–150. [\[CrossRef\]](#)
37. Lee, J.; Davari, H.; Singh, J.; Pandhare, V. Industrial Artificial Intelligence for industry 4.0-based manufacturing systems. *Manuf. Lett.* **2018**, *18*, 20–23. [\[CrossRef\]](#)
38. Gabriel, M.; Pessl, E. Industry 4.0 and sustainability impacts: Critical discussion of sustainability aspects with a special focus on the future of work and ecological consequences. *Ann. Fac. Eng. Hunedoara* **2016**, *14*, 131–136.
39. Willard, B. *The New Sustainability Advantage: Seven Business Case Benefits of a Triple Bottom Line*/Bob Willard; Completely rev. 10th anniversary ed.; New Society Publishers: Gabriola Island, BC, USA, 2012; ISBN 0865717125.
40. Jeske, T.; Weber, M.-A.; Würfels, M.; Lennings, F.; Stowasser, S. Opportunities of Digitalization for Productivity Management. In *Advances in Human Factors and Systems Interaction*; Nunes, I.L., Ed.; Springer International Publishing: Cham, Switzerland, 2019; pp. 321–331, ISBN 978-3-319-94333-6.
41. Stock, T.; Seliger, G. Opportunities of Sustainable Manufacturing in Industry 4.0. *Procedia CIRP* **2016**, *40*, 536–541. [\[CrossRef\]](#)

42. de Sousa Jabbour, A.B.L.; Jabbour, C.J.C.; Foropon, C.; Godinho Filho, M. When titans meet—Can industry 4.0 revolutionise the environmentally-sustainable manufacturing wave? The role of critical success factors. *Technol. Forecast. Soc. Chang.* **2018**, *132*, 18–25. [CrossRef]
43. Cochran, D.S.; Rauch, E. Sustainable Enterprise Design 4.0: Addressing Industry 4.0 Technologies from the Perspective of Sustainability. *Procedia Manuf.* **2020**, *51*, 1237–1244. [CrossRef]
44. Cozmiuc, D.; Petrisor, I. Industrie 4.0 by Siemens. *J. Cases Inf. Technol.* **2018**, *20*, 30–48. [CrossRef]
45. Blackstock, M.; Lea, R. IoT interoperability: A hub-based approach. In Proceedings of the 2014 International Conference on the Internet of Things (IOT), Cambridge, MA, USA, 6–8 October 2014; IEEE: New York, NY, USA, 2014; pp. 79–84, ISBN 978-1-4799-5154-3.
46. Miller, A.M.; Alvarez, R.; Hartman, N. Towards an extended model-based definition for the digital twin. *Comput.-Aided Des. Appl.* **2018**, *15*, 880–891. [CrossRef]
47. Hold, P.; Erol, S.; Reisinger, G.; Sihm, W. Planning and Evaluation of Digital Assistance Systems. *Procedia Manuf.* **2017**, *9*, 143–150. [CrossRef]
48. Marcon, P.; Zezulka, F.; Vesely, I.; Szabo, Z.; Roubal, Z.; Sajdl, O.; Gescheidtova, E.; Dohnal, P. Communication technology for industry 4.0. In Proceedings of the 2017 Progress in Electromagnetics Research Symposium-Spring (PIERS), St. Petersburg, Russia, 22–25 May 2017; IEEE: New York, NY, USA, 2017; pp. 1694–1697, ISBN 978-1-5090-6269-0.
49. Peruzzini, M.; Grandi, F.; Pellicciari, M. Benchmarking of Tools for User Experience Analysis in Industry 4.0. *Procedia Manuf.* **2017**, *11*, 806–813. [CrossRef]
50. Ceruti, A.; Marzocca, P.; Liverani, A.; Bil, C. Maintenance in aeronautics in an Industry 4.0 context: The role of Augmented Reality and Additive Manufacturing. *J. Comput. Des. Eng.* **2019**, *6*, 516–526. [CrossRef]
51. Santos, M.Y.; Oliveira e Sá, J.; Andrade, C.; Vale Lima, F.; Costa, E.; Costa, C.; Martinho, B.; Galvão, J. A Big Data system supporting Bosch Braga Industry 4.0 strategy. *Int. J. Inf. Manag.* **2017**, *37*, 750–760. [CrossRef]
52. Zheng, T.; Ardolino, M.; Bacchetti, A.; Perona, M. The applications of Industry 4.0 technologies in manufacturing context: A systematic literature review. *Int. J. Prod. Res.* **2021**, *59*, 1922–1954. [CrossRef]
53. Federal Aviation Administration. *Aircraft Inspection and Maintenance Records*; Federal Aviation Administration: Washington, DC, USA, 2018; ISBN 0884873196.
54. French, R.; Marin-Reyes, H.; Benakis, M. Transfer Analysis of Human Engineering Skills for Adaptive Robotic Additive Manufacturing in the Aerospace Repair and Overhaul Industry. In *Advances in Manufacturing, Production Management and Process Control*; Karwowski, W., Trzcielinski, S., Mrugalska, B., Di Nicolantonio, M., Rossi, E., Eds.; Springer International Publishing: Cham, Switzerland, 2019; pp. 3–12, ISBN 978-3-319-94195-0.
55. Horonjeff, R.; McKelvey, F.X.; Sproule, W.J.; Young, S.B. *Planning and Design of Airports, 5th Edition*; McGraw-Hill Education: New York, NY, USA, 2010; ISBN 9780071446419.
56. Dalkiran, A.; Ayar, M.; Kale, U.; Nagy, A.; Karakoc, T.H. A review on thematic and chronological framework of impact assessment for green airports. *Int. J. Green Energy* **2022**, 1–12. [CrossRef]
57. Xu, J.K.; Sun, S.M. The Current Situation, Opportunities and Challenges of Green Construction and Sustainable Development in Chinese Airports. *Adv. Mater. Res.* **2013**, *860–863*, 1289–1296. [CrossRef]
58. Wang, Z.-J.; Jia, H.-H.; Dai, F.; Diao, M. Understanding the ground access and airport choice behavior of air passengers using transit payment transaction data. *Transp. Policy* **2022**, *127*, 179–190. [CrossRef]
59. Sun, L.; Pan, H.; Hu, X. Short review of concepts and practices in green airports in China. *J. Phys. Conf. Ser.* **2021**, *1976*, 12063. [CrossRef]
60. Wang, X.; Lv, M. China Aviation Electronic Miscellaneous Document System. In *Advances in Electrical Engineering and Electrical Machines*; Zheng, D., Ed.; Springer Berlin Heidelberg: Berlin/Heidelberg, Germany, 2011; pp. 167–172, ISBN 978-3-642-25904-3.
61. Liu, R.; Zou, Y.; Chen, X.; Yang, H. Electronic Aircraft Maintenance Record Signature Technology Based on SM2 Cryptographic Algorithm. In Proceedings of the 2019 IEEE 1st International Conference on Civil Aviation Safety and Information Technology (ICCASIT), Kunming, China, 17–19 October 2019.
62. Bisignani, G. Director General & CEO International Air Transport Association Annual Report. In Proceedings of the 67th Annual General Meeting, Singapore, 2011.
63. López-Bonilla, J.M.; López-Bonilla, L.M. Self-Service Technology Versus Traditional Service: Examining Cognitive Factors in The Purchase of The Airline Ticket. *J. Travel Tour. Mark.* **2013**, *30*, 497–508. [CrossRef]
64. Backes, M.; Miwa, F.; Okajima, C.; Souza, A., Jr.; Tkacz, D. From Paper to Digital Maintenance with Electronic Signature. 2017. Available online: https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwiSjcStkYuCAxXVbvUHHQH-C0oQFnoECA4QAQ&url=https%3A%2F%2Fcommons.erau.edu%2Fcgi%2Fviewcontent.cgi%3Farticle%3D1015%26context%3Dbrazil-graduate-works&usg=AOvVaw0_Y4HddY6-LFoW_pNfZT5O&opi=89978449 (accessed on 12 June 2020).
65. MacKay, W.E. Is paper safer? The role of paper flight strips in air traffic control. *ACM Trans. Comput.-Hum. Interact.* **1999**, *6*, 311–340. [CrossRef]
66. Hurter, C.; Lesbordes, R.; Letondal, C.; Vinot, J.-L.; Conversy, S. StripTIC. In Proceedings of the AVI'12: International Working Conference on Advanced Visual Interfaces, Capri Island, Italy, 21–25 May 2012; Tortora, G., Levialdi, S., Tucci, M., Eds.; ACM: New York, NY, USA, 2012; pp. 225–232, ISBN 9781450312875.

67. Fields, R.; Marti, P.; Termini, V.; Palmonari, M. Air Traffic Control as a Distributed Cognitive System: A study of external representations. In Proceedings of the ECCE9: Proceedings of the Ninth European Conference on Cognitive Ergonomics, Limerick, Ireland, 24–26 August 1998.
68. Hilburn, B. Head-down time in ATC tower operations: Real time simulations results. *Cent. Hum. Perform. Res.* **2004**.
69. Bos, T.; Schuwer-van Blanken, M.; Huisman, H. Towards a Paperless Air Traffic Control Tower. In *Human Centered Design*; Kurosu, M., Ed.; Springer Berlin Heidelberg: Berlin/Heidelberg, Germany, 2011; pp. 360–368, ISBN 978-3-642-21752-4.
70. Durso, F.T.; Manning, C.A. *Human Factors and Aerospace Safety: SPINNING PAPER INTO GLASS: TRANSFORMING FLIGHT PROGRESS STRIPS*, 1st ed.; Ashgate Publishing: Farnham, UK, 2002.
71. Pavet, D. *Use of Paper Strips by Tower Air Traffic Controllers and Promises Offered by New Design Techniques on User Interface*; USA/Europe R&D Seminar: Santa-Fe, NM, USA, 2001.
72. Edwards, M.B.; Fuller, D.K.; Vortac, O.U.; Manning, C.A. The role of flight progress strips in en route air traffic control: A time-series analysis. *Int. J. Hum.-Comput. Stud.* **1995**, *43*, 1–13. [\[CrossRef\]](#)
73. Seinfeld, R.D.; Herman, R.P.; Lotterhos, L. Pilot information display for the paperless cockpit, and more. In *Cockpit Displays IX: Displays for Defense Applications, Proceedings of the AeroSense 2002, Orlando, FL, USA, 1 April 2002*; Hopper, D.G., Ed.; SPIE: Bellingham, WA, USA, 2002; pp. 8–13.
74. Fitzsimmons, F. *The Electronic Flight Bag: A Multi-Function Tool for the Modern Cockpit*; Institute for Information Technology Applications: Onnason, Japan, 2002; p. 66.
75. Takahashi, T. Ipad's in the Cockpit: Evolution or Revolution in the Sky. *SSRN J.* **2012**. [\[CrossRef\]](#)
76. Quinlan, M.; Hampson, I.; Gregson, S. Outsourcing and offshoring aircraft maintenance in the US: Implications for safety. *Saf. Sci.* **2013**, *57*, 283–292. [\[CrossRef\]](#)
77. Chang, Y.; Oh, C.; Whang, Y.; Lee, J.; Kwon, J.; Kang, M.; Park, J.S.; Park, U. Development of RFID Enabled Aircraft Maintenance System. In Proceedings of the 2006 IEEE International Conference on Industrial Informatics, Singapore, 16–18 August 2006; IEEE: New York, NY, USA, 2006; pp. 224–229, ISBN 0-7803-9701-0.
78. Tsakalerou, M.; Nurmaganbetov, D.; Beltenov, N. Aircraft Maintenance 4.0 in an era of disruptions. *Procedia Comput. Sci.* **2022**, *200*, 121–131. [\[CrossRef\]](#)
79. Elakramine, F.; Jaradat, R.; Ullah Ibne Hossain, N.; Banghart, M.; Kerr, C.; El Amrani, S. Applying Systems Modeling Language in an Aviation Maintenance System. *IEEE Trans. Eng. Manag.* **2022**, *69*, 4006–4018. [\[CrossRef\]](#)
80. Mackenzie, D. *ICAO: A History of the International Civil Aviation Organization*; University of Toronto Press: Toronto, ON, Canada, 2010; ISBN 9781442640108.
81. ICAO. Guidance For Acceptance of Electronic Aircraft Maintenance Records (EAMR). Available online: <https://www.icao.int/safety/airnavigation/OPS/airworthiness/Pages/EAMR.aspx> (accessed on 12 June 2020).
82. IATA. *Guidance Material for the Implementation of Paperless Aircraft Operations in Technical Operations (PAO:TO)*; International Air Transport Association: Montreal, QC, Canada, 2017.
83. Cunliffe, A.M.; Anderson, K.; DeBell, L.; Duffy, J.P. A UK Civil Aviation Authority (CAA)-approved operations manual for safe deployment of lightweight drones in research. *Int. J. Remote Sens.* **2017**, *38*, 2737–2744. [\[CrossRef\]](#)
84. Kivanç, E.; Vayvay, Ö.; Kalender, Z.T. Digital Transformation Approaches for Aircraft Maintenance Operations. In *Managerial Issues in Digital Transformation of Global Modern Corporations*; Wang, J., Esakki, T., Eds.; IGI Global: Hershey, PA, USA, 2021; pp. 145–164, ISBN 9781799824022.
85. IATA. Aviation Identification & Authorisation System (White Paper). 2015. Available online: https://www.iata.org/contentassets/fafa409c883d41198aeb87628c848851/aviation_identification_20authorisation_system_-_white_paper_-_v1_2015.pdf (accessed on 18 June 2020).
86. Scott, T. Digital Aircraft and Engine Lease Returns (White Paper). 2016. Available online: <https://www.aircraftit.com/articles/digital-aircraftand-engine-lease-returns/> (accessed on 12 August 2023).
87. IATA. Guidance Material and Best Practices for Aircraft Leases. 2017. Available online: <https://www.iata.org/en/programs/workgroups/aircraft-leasing> (accessed on 12 June 2020).
88. Walker, D.; Laczniak, R.N.; Carlson, L.; Brocato, E.D. Parenting Orientations as Antecedents of Children's Violent Videogame Play. *J. Consum. Aff.* **2016**, *50*, 430–457. [\[CrossRef\]](#)
89. Hensel, J. *Case Study: Jazz Aviation—Paperless Maintenance Processes*; Aircraft IT MRO: Bangkok, Thailand, 2017.
90. Scott, T. (Ed.) *Digital Aircraft and Engine Lease Returns (White Paper)*; AVITAS: Washington, DC, USA, 2016.
91. Aleshi, A.; Seker, R.; Babiceanu, R.F. Blockchain Model for Enhancing Aircraft Maintenance Records Security. In Proceedings of the 2019 IEEE International Symposium on Technologies for Homeland Security (HST), Woburn, MA, USA, 5–6 November 2019; IEEE: New York, NY, USA, 2019; pp. 1–7, ISBN 978-1-7281-5092-5.
92. Federal Aviation Administration. *Aviation Maintenance Technician Handbook—General ([Edition Unavailable])*; Federal Aviation Administration: Washington, DC, USA, 2018.
93. Kozolchyk, B.; Bayitch, S.A. Aircraft Mortgage in the Americas. A Study in Comparative Aviation Law with Documents. *Am. J. Comp. Law* **1961**, *10*, 287. [\[CrossRef\]](#)
94. Wirsamulia, F. The Legal Problem of Aircraft Mortgage in Indonesia. *J. Leg. Ethical Regul. Issues* **2021**, *24*, 1–10.

95. Shoshin, L.; Susanin, V. Digital Transformation of an Aircraft Operation Ecosystem. In *Reliability and Statistics in Transportation and Communication*; Kabashkin, I., Yatskiv, I., Prentkovskis, O., Eds.; Springer International Publishing: Cham, Switzerland, 2022; pp. 198–212, ISBN 978-3-030-96195-4.
96. Prasetyo, S.E.; Damaraji, G.M.; Kusumawardani, S.S. A Review of The Challenges of Paperless Concept in The Society 5.0. *Int. J. Ind. Eng. Eng. Manag.* **2020**, *2*, 15–24. [[CrossRef](#)]
97. Belobaba, P.; Swelbar, W.; Barnhart, C. Information Technology in Airline Operations, Distribution and Passenger Processing. In *The Global Airline Industry*; Belobaba, P., Odoni, A., Barnhart, C., Eds.; John Wiley & Sons, Ltd.: Chichester, UK, 2009; pp. 441–466, ISBN 9780470744734.
98. IBA. IBA's Redelivery Whitepaper: The Picture in 2018. 2018. Available online: <https://www.iba.aero/insight/iba-redelivery-whitepaper-the-picture-in-2018/> (accessed on 12 August 2023).
99. Ackert, S. Redelivery Considerations in Aircraft Operating Leases: Guidelines and Best Practices to Ease Transferability of Aircraft. 2014. Available online: http://www.aircraftmonitor.com/uploads/1/5/9/9/15993320/redelivery_considerations_in_acft_op_leases__v1.pdf (accessed on 15 August 2023).
100. Shanmugam, A.; Paul Robert, T. Human factors engineering in aircraft maintenance: A review. *J. Qual. Maint. Eng.* **2015**, *21*, 478–505. [[CrossRef](#)]
101. Junqueira, V.S.V.; Nagano, M.S.; Miyata, H.H. Procedure structuring for programming aircraft maintenance activities. *REGE* **2020**, *27*, 2–20. [[CrossRef](#)]
102. Endsley, M.R.; Robertson, M.M. Situation awareness in aircraft maintenance teams. *Int. J. Ind. Ergon.* **2000**, *26*, 301–325. [[CrossRef](#)]
103. PeriyarSelvam, U.; Tamilselvan, T.; Thilakan, S.; Shanmugaraja, M. Analysis on Costs for Aircraft Maintenance. *Adv. Aerosp. Sci. Appl.* **2013**, *3*, 177–182.

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